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15 October 1968
SUPERSEDING
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19 February 1963

MILITARY SPECIFICATION

AREA PREDICTOR, RADIOLOGICAL FALLOUT, M5A1

1. SCOPE

1.1 This specification covers a radiological fallout area predictor, which is a transparent vinyl overlay for a map, and its case.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

MM-P-530	-	Plywood, Flat-Panel.
QQ-S-781	-	Steel, Strapping, Flat.
TT-W-572	-	Wood-Preservative; Water-Repellent.
UU-T-106	-	Tape, Pressure-Sensitive Adhesive, Masking, Paper.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-591	-	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-B-22205	-	Bags, Transparent, Flexible, Heat Sealable For Packaging Applications.

FSC 6665

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL124-13-24 - Area Predictor, Radiological Fallout, M5A1.
A18-17-14 - Insignia, CML.C., 3 1/2.
D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials cited on Drawing DL124-13-24 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings. The ink used for marking and lettering shall be compatible with the vinyl chloride material. The marking and lettering shall be legible, non-fading, and durable.

3.2 Manufacture.

3.2.1 Predictor. The predictor shall be manufactured in accordance with Drawing E124-13-24.

3.2.2 Case. The case shall be manufactured in accordance with Drawing D124-13-26.

3.3 Assembly. The fallout predictor shall be folded and assembled into the case.

3.4 Functioning. The predictor, when tested in accordance with 4.4.4.1, shall be capable of solving problems for downwind distance using the data listed in table I.

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Table I. Problem data

Scale	Data			
	Problem 1	Problem 2	Problem 3	Problem 4
Yield - KT	1,000	400	20	0.2
Effective wind speed - KM/Hr.	55	21	20	40
Downwind distance - KM	150 ± 5	60 ± 3	18 ± 1	3.6 ± 0.2

3.5 Marking stability. The stability of the marking on the fallout area predictor shall be such that, when tested as specified in 4.4.4.2, not more than three characters (letters and numbers) shall have breaks or parts missing; not more than two breaks shall be present in any circle, semi-circle, or distance scale; not more than two tick marks on the azimuth or distance scale shall be partially obliterated; and no single tick mark on the azimuth or distance scale shall be completely missing.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of M5A1 area predictors shall be produced in accordance with this specification for examination and test (see 4.3).

3.7 Workmanship. The predictor and case shall be free from imperfections such as cuts, tears, punctures, cracks, blisters, bumps or dents and contamination such as grease, oil or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five M5A1 fallout area predictors shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedures for examination and nondestructive tests. The preproduction sample of area predictors shall be examined for all the requirements of the drawings and this specification and shall be tested as specified in 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the fallout area predictors produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For Major defect 107 level S-4 shall be used, and for Major defect 108 level S-2 shall be used.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample area predictors and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

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4.4.3.2 Classification of defects.

(a) Area predictor, radiological fallout, M5A1 (Dwg E124-13-24).

Categories	Defects	Acceptance standard
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Marking or lettering incorrect, illegible, or missing	
102	Color of markings	
103	Lettering and marking not on underneath side	
104	Overall length	
105	Overall width	
106	Workmanship (see 3.7)	
107	Functioning	4.4.4.1
	AQL 2.5 percent defective	
108	Marking stability	4.4.4.2

(b) Case (Dwg D124-13-26).

Categories	Defects
Critical:	None defined
Major:	AQL 1.5 percent defective
101	Overall length
102	Overall width
103	Heat seals not continuous
104	Marking missing, incorrect, illegible or incorrectly located
105	Workmanship (see 3.7)

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(c) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 6.5 percent defective
101	Unit packaging incorrect or missing
102	Quantity per container incorrect, when specified
103	Packaging or packing container incorrect
104	Container broken, torn, crushed or punctured
105	Marking missing, incorrect or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning. From the nomogram on the area predictor, solve the four problems using the data listed in table I. Align the yield data and effective wind speed with a straight edge. Read the downwind distance from ground zero on the middle scale and compare with table I.

4.4.4.2 Marking stability. Submerge the area predictor in a suitable container of water maintained at 125° ± 5° F for 30 ± 2 minutes with the entire printed side of the predictor in contact with the water. Remove the predictor from the water and wipe dry. Place the predictor on a smooth hard surface with the printed side of the predictor up. Within four minutes after removal from the water, apply five 1/2-inch by 6-inch strips of masking tape, conforming to type II of UU-T-106, at right angles to the respective scales and nomograph so that portions of both the numbers and letters are covered. Roll a 10-pound, 5-inch diameter, hard surfaced roller at approximately 12 inches per 10 seconds over the strips of tape three times. Double back the end of each strip of tape and slowly and evenly peel each strip from the area predictor.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level shall be as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each fallout area predictor, in its case, shall be sealed in a bag conforming to type II, size 7 of MIL-B-22205. Included air volume shall be kept to a practicable minimum.

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5.1.1.2 Intermediate packaging. Fallout area predictors, unit packaged in accordance with 5.1.1.1, shall be intermediate packaged, in even unit quantities or quantities specified in the contract or order (see 6.2), in appropriate size boxes conforming to type CF, class weather-resistant, variety SW of PPP-B-636. The quantity per box shall not exceed the size and weight limitations for the respective type box in accordance with PPP-B-636. The unit packages shall be restricted from free movement and the box closed by gluing, steel stitching, or stapling as required in the appendix of PPP-B-636 for weather-resistant boxes.

5.1.2 Level C. Fallout area predictors, in their cases, shall be packaged in even unit quantities or quantities as specified in the contract or order in fiberboard boxes conforming to class domestic of PPP-B-636. The quantity per box shall not exceed the size and weight limitations for the respective box in accordance with PPP-B-636. The items shall be restricted from free movement and the box closed in accordance with method II of the appendix to PPP-B-636.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. Fallout area predictors, packaged as specified in 5.1.1, shall be packed in even unit quantities or quantities specified in the contract or order in boxes conforming to either class 2, style 4 of PPP-B-621; style A, overseas type of PPP-B-601; or class 3, style 2 or 3 of PPP-B-585 (see 6.2). Boxes shall be preservative treated by completely immersing the boxes or wooden box components for a minimum period of 3 minutes in a water-repellent wood preservative conforming to type II, composition A of TT-W-572. Also, when plywood material is utilized, it shall conform to group A, CS 35, type I of NN-P-530, and shall be free of holes or openings extending through from the face to the back. Boxes shall fit the contents fairly snug. Filler pads of fiberboard material conforming to weather-resistant class, grade optional of PPP-F-320, shall be included, as necessary, to restrict free movement of the packages and the boxes shall be securely closed and strapped in accordance with the respective box specification and the appendix thereto. Strapping shall conform to class B, type I or IV of QQ-S-781.

5.2.2 Level B. Fallout area predictors, packaged as specified, shall be packed in accordance with 5.2.1 except boxes may be domestic class and preservative treatment is not required.

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5.2.3 Level C. Fallout area predictors, packaged as specified shall be packed in even unit quantities or quantities specified in the contract or order in fiberboard or wood boxes conforming to either weather-resistant class of PPP-B-636; class 1 of PPP-B-585;; class 1 of PPP-B-591; domestic type of PPP-B-601; or class 1 of PPP-B-621 (see 6.2). The packages shall be restricted from free movement and the boxes shall be securely closed and strapped in accordance with the respective box specification and the appendix thereto. Unless otherwise specified in the contract or order, fiberboard boxes shall be closed by gluing in accordance with requirements for weather-resistant class in the appendix to PPP-B-636.

5.3 Marking In conjunction with pertinent information provided in the contract or order, including special marking requirements, packages shall be marked in accordance with MIL-STD-129. Exterior shipping containers shall be marked in accordance with Drawing D150-2-1. Included as a special marking, as applicable (see 6.2), shall be a crossed retorts insignia conforming to Drawing A18-17-14 applied on the exterior container in an otherwise unmarked area adjacent to the identification marking.

6. NOTES

6.1 Intended use. This predictor is intended for use in making an immediate estimate of the potential location of the radioactive fallout hazard following a nuclear burst.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Number of fallout area predictors per intermediate container required.
(d) Type of exterior container or optional.
(e) Crossed retorts insignia (Dwg A18-17-14) for level A packs.

Custodian

Army - MU

Preparing activity:

Army - MU(EA)

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